Task 4: Order Management using Joins:

Aim

To design and implement an order management system using SQL joins and subqueries to retrieve customer order history, identify the highest value order, and determine the most active customer.

Algorithm

1. **Start**
2. **Create Tables**
    - **Create Customers table (CustomerID, Name, Email)**
    - **Create Products table (ProductID, ProductName, Price)**
    - **Create Orders table (OrderID, CustomerID, ProductID, Quantity, OrderDate)**
3. **Insert Data**
    - **Add sample records into all three tables.**
4. **Apply JOIN Query**
    - **Use INNER JOIN to combine:**
        - **Customers + Orders + Products**
    - **Display:**
        - **Customer Name**
        - **Product Name**
        - **Quantity**
        - **Total Price (Quantity × Price)**
        - **Order Date**
5. **Find Highest Value Order (Subquery)**
    - **Calculate total price for each order.**
    - **Use subquery to find maximum order value.**
6. **Find Most Active Customer (Subquery)**

- Count number of orders per customer.
- Use subquery to find the customer with the highest count.

7. Sort Results
   - Use ORDER BY to arrange records (e.g., by date or total amount).
8. Display Output
   - Show results in tabular format (optionally styled with CSS).
9. End

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Order Management SQL Dashboard</title>
  <script src="https://cdn.tailwindcss.com"></script>
  <style>
    :root {
      --primary: #4f46e5;
      --secondary: #0ea5e9;
      --bg: #f8fafc;
    }
    body {
      background-color: var(--bg);
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
    }
    .sql-card {
      background: white;
```

```
      border-radius: 12px;
      box-shadow: 0 4px 6px -1px rgb(0 0 0 / 0.1);
      transition: transform 0.2s;
    }
    .sql-card:hover {
      transform: translateY(-2px);
    }
    .query-box {
      background: #1e293b;
      color: #e2e8f0;
      padding: 1rem;
      border-radius: 8px;
      font-family: 'Fira Code', monospace;
      font-size: 0.9rem;
      overflow-x: auto;
    }
    .keyword { color: #f472b6; font-weight: bold; }
    .function { color: #38bdf8; }
    .string { color: #fbbf24; }
    .table-name { color: #4ade80; }
    .badge {
      padding: 2px 8px;
      border-radius: 9999px;
      font-size: 0.75rem;
      font-weight: 600;
    }
  </style>
</head>
```

```
<body class="p-4 md:p-8">
  <div class="max-w-6xl mx-auto">
    <header class="mb-8 border-b border-slate-200 pb-6">
      <h1 class="text-3xl font-bold text-slate-900">Retail Order Management System</h1>
      <p class="text-slate-600">Analyzing relational data with JOINS, Subqueries, and Complex Ordering.</p>
    </header>


    <!-- Database Schema Overview -->
    <div class="grid grid-cols-1 md:grid-cols-3 gap-6 mb-10">
      <!-- Customers Table -->
      <div class="sql-card p-4">
        <div class="flex items-center justify-between mb-4">
          <h3 class="font-bold text-slate-800 flex items-center gap-2">
            <span class="w-3 h-3 bg-indigo-500 rounded-full"></span> Customers
          </h3>
          <span class="text-xs text-slate-400">TABLE</span>
        </div>
        <div class="overflow-hidden rounded border border-slate-100">
          <table class="w-full text-left text-xs">
            <thead class="bg-slate-50 text-slate-500">
              <tr>
                <th class="p-2 border-b">id</th>
                <th class="p-2 border-b">name</th>
                <th class="p-2 border-b">email</th>
              </tr>
            </thead>
            <tbody class="text-slate-700" id="table-customers"></tbody>
```

```
      </table>
    </div>
  </div>

  <!-- Products Table -->
  <div class="sql-card p-4">
    <div class="flex items-center justify-between mb-4">
      <h3 class="font-bold text-slate-800 flex items-center gap-2">
        <span class="w-3 h-3 bg-emerald-500 rounded-full"></span> Products
      </h3>
      <span class="text-xs text-slate-400">TABLE</span>
    </div>
    <div class="overflow-hidden rounded border border-slate-100">
      <table class="w-full text-left text-xs">
        <thead class="bg-slate-50 text-slate-500">
          <tr>
            <th class="p-2 border-b">id</th>
            <th class="p-2 border-b">name</th>
            <th class="p-2 border-b">price</th>
          </tr>
        </thead>
        <tbody class="text-slate-700" id="table-products"></tbody>
      </table>
    </div>
  </div>

  <!-- Orders Table -->
  <div class="sql-card p-4">
```

```
      <div class="flex items-center justify-between mb-4">
        <h3 class="font-bold text-slate-800 flex items-center gap-2">
          <span class="w-3 h-3 bg-amber-500 rounded-full"></span> Orders
        </h3>
        <span class="text-xs text-slate-400">TABLE</span>
      </div>
      <div class="overflow-hidden rounded border border-slate-100">
        <table class="w-full text-left text-xs">
          <thead class="bg-slate-50 text-slate-500">
            <tr>
              <th class="p-2 border-b">id</th>
              <th class="p-2 border-b">cust_id</th>
              <th class="p-2 border-b">prod_id</th>
              <th class="p-2 border-b">qty</th>
            </tr>
          </thead>
          <tbody class="text-slate-700" id="table-orders"></tbody>
        </table>
      </div>
    </div>
  </div>

  <!-- Analytical Section -->
  <div class="grid grid-cols-1 lg:grid-cols-2 gap-8">
    <!-- Left Side: Order History JOIN -->
    <div>
      <div class="flex items-center gap-3 mb-4">
        <h2 class="text-xl font-bold text-slate-800">Master Order History</h2>
```

```
        <span class="bg-indigo-100 text-indigo-700 text-xs px-2 py-1 rounded">INNER JOIN</span>
      </div>

      <div class="query-box mb-4">
        <span class="keyword">SELECT</span> o.id, c.name, p.product_name, (o.qty * p.price) <span class="keyword">AS</span> total<br>
        <span class="keyword">FROM</span> orders o<br>
        <span class="keyword">JOIN</span> customers c <span class="keyword">ON</span> o.customer_id = c.id<br>
        <span class="keyword">JOIN</span> products p <span class="keyword">ON</span> o.product_id = p.id<br>
        <span class="keyword">ORDER BY</span> total <span class="keyword">DESC</span>;
      </div>

      <div class="bg-white rounded-xl shadow-sm border border-slate-200 overflow-hidden">
        <table class="w-full text-sm text-left">
          <thead class="bg-slate-50 text-slate-600 font-medium">
            <tr>
              <th class="px-4 py-3 border-b">Order ID</th>
              <th class="px-4 py-3 border-b">Customer</th>
              <th class="px-4 py-3 border-b">Product</th>
              <th class="px-4 py-3 border-b text-right">Order Value</th>
            </tr>
          </thead>
          <tbody id="join-results" class="divide-y divide-slate-100"></tbody>
        </table>
      </div>
```

```
    </div>

    <!-- Right Side: Subqueries and Insights -->
    <div class="space-y-6">
      <!-- Most Active Customer -->
      <div>
        <h2 class="text-xl font-bold text-slate-800 mb-4 flex items-center gap-2">
          <span>Customer Insights</span>
          <span class="text-xs font-normal text-slate-400">SUBQUERIES</span>
        </h2>

        <div class="grid grid-cols-1 gap-4">
          <!-- Highest Value Order Card -->
          <div class="bg-white p-5 rounded-xl border border-slate-200 shadow-
sm">
            <p class="text-slate-500 text-sm font-medium mb-1">Highest Value
Order</p>
            <div class="flex items-end justify-between">
              <h3 id="stat-high-val" class="text-2xl font-bold text-indigo-
600">$0.00</h3>
              <div class="text-right">
                <p id="stat-high-name" class="text-xs font-semibold text-slate-
700">Loading...</p>
                <p id="stat-high-product" class="text-[10px] text-slate-
400">Loading...</p>
              </div>
            </div>
            <div class="mt-4 text-[10px] query-box opacity-75">
              <span class="keyword">SELECT</span> * <span
class="keyword">FROM</span> orders <span class="keyword">WHERE</span>
```

```
total = (<span class="keyword">SELECT</span> <span
class="function">MAX</span>(total) <span class="keyword">FROM</span>
orders)
                </div>
              </div>


              <!-- Most Active Customer Card -->
              <div class="bg-white p-5 rounded-xl border border-slate-200 shadow-
sm">
                <p class="text-slate-500 text-sm font-medium mb-1">Most Active
Customer</p>
                <div class="flex items-end justify-between">
                  <h3 id="stat-active-cust" class="text-2xl font-bold text-emerald-
600">User</h3>
                  <div class="text-right">
                    <p id="stat-active-count" class="text-xs font-semibold text-slate-
700">0 Orders</p>
                    <p class="text-[10px] text-slate-400">Retention Leader</p>
                  </div>
                </div>
              </div>
            </div>
          </div>


          <!-- Footer Summary Info -->
          <div class="bg-slate-100 p-4 rounded-lg text-sm text-slate-600 italic">
            <p>Real-time application: In e-commerce, these JOINs allow us to bridge
the gap between "Who the person is" and "What they bought" by connecting IDs
across table boundaries.</p>
          </div>
```

```
      </div>
    </div>
  </div>

<script>
  // Mock Database Data
  const customers = [
    { id: 101, name: "Alice Johnson", email: "alice@example.com" },
    { id: 102, name: "Bob Smith", email: "bob@tech.org" },
    { id: 103, name: "Charlie Davis", email: "charlie@gmail.com" },
    { id: 104, name: "Diana Prince", email: "diana@amazon.com" }
  ];

  const products = [
    { id: 501, name: "UltraBook Pro", price: 1200 },
    { id: 502, name: "Noise-Cancelling Headphones", price: 350 },
    { id: 503, name: "Mechanical Keyboard", price: 150 },
    { id: 504, name: "4K Monitor 27\"", price: 450 }
  ];

  const orders = [
    { id: 1, cust_id: 101, prod_id: 501, qty: 1 },
    { id: 2, cust_id: 102, prod_id: 503, qty: 2 },
    { id: 3, cust_id: 101, prod_id: 502, qty: 1 },
    { id: 4, cust_id: 104, prod_id: 504, qty: 3 },
    { id: 5, cust_id: 103, prod_id: 502, qty: 1 },
    { id: 6, cust_id: 101, prod_id: 503, qty: 1 },
    { id: 7, cust_id: 102, prod_id: 501, qty: 1 }
```

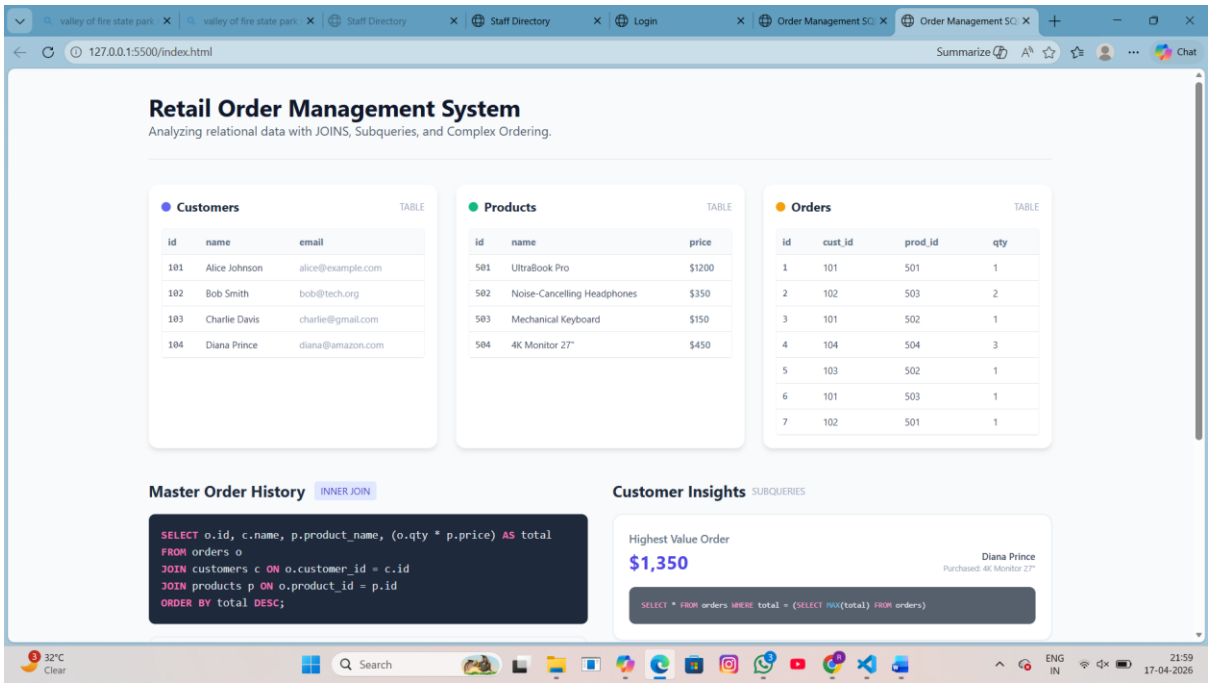
```
];

function init() {
  renderBaseTables();
  renderJoinResults();
  calculateInsights();
}

function renderBaseTables() {
  const custTbody = document.getElementById('table-customers');
  customers.forEach(c => {
    custTbody.innerHTML += `<tr><td class="p-2 border-b font-mono">${c.id}</td><td class="p-2 border-b">${c.name}</td><td class="p-2 border-b text-slate-400">${c.email}</td></tr>`;
  });

  const prodTbody = document.getElementById('table-products');
  products.forEach(p => {
    prodTbody.innerHTML += `<tr><td class="p-2 border-b font-mono">${p.id}</td><td class="p-2 border-b">${p.name}</td><td class="p-2 border-b text-slate-500 font-semibold">$${p.price}</td></tr>`;
  });

  const orderTbody = document.getElementById('table-orders');
  orders.forEach(o => {
    orderTbody.innerHTML += `<tr><td class="p-2 border-b font-mono">${o.id}</td><td class="p-2 border-b">${o.cust_id}</td><td class="p-2 border-b">${o.prod_id}</td><td class="p-2 border-b">${o.qty}</td></tr>`;
  });
}
```

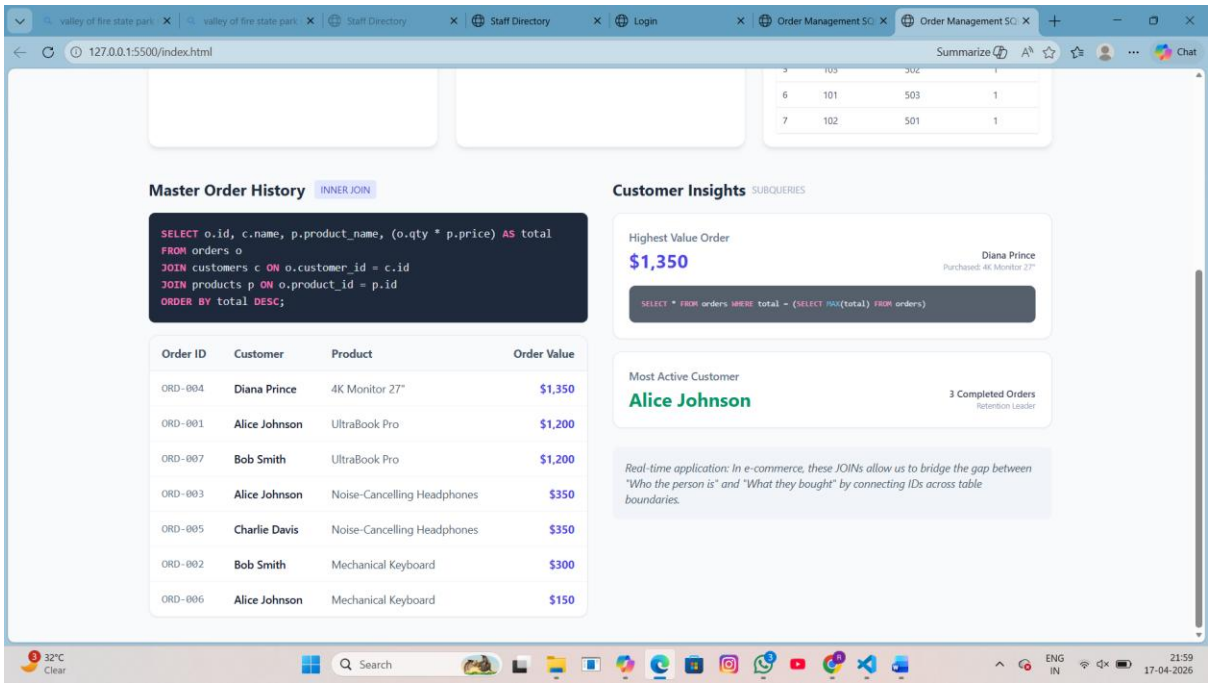
```
function renderJoinResults() {
  const tbody = document.getElementById('join-results');

  // Simulating SQL:
  // JOIN orders ON customers (cust_id)
  // JOIN products ON orders (prod_id)
  const results = orders.map(o => {
    const customer = customers.find(c => c.id === o.cust_id);
    const product = products.find(p => p.id === o.prod_id);
    const total = o.qty * product.price;
    return {
      id: o.id,
      name: customer.name,
      product: product.name,
      total
    };
  }).sort((a, b) => b.total - a.total); // ORDER BY total DESC

  results.forEach(res => {
    tbody.innerHTML += `
      <tr class="hover:bg-slate-50">
        <td class="px-4 py-3 font-mono text-slate-500">ORD-00${res.id}</td>
        <td class="px-4 py-3 font-medium text-slate-900">${res.name}</td>
        <td class="px-4 py-3 text-slate-600">${res.product}</td>
        <td class="px-4 py-3 text-right font-bold text-indigo-
600">$${res.total.toLocaleString()}</td>
      </tr>
```

```
        `;
    });
  }

  function calculateInsights() {
    // Calculating "Highest Value Order" using logic similar to Subquery
    const orderValues = orders.map(o => {
      const p = products.find(prod => prod.id === o.prod_id);
      return { ...o, total: o.qty * p.price, productName: p.name };
    });

    const maxOrder = orderValues.reduce((prev, current) => (prev.total >
current.total) ? prev : current);
    const maxCust = customers.find(c => c.id === maxOrder.cust_id);

    document.getElementById('stat-high-val').innerText =
`$${maxOrder.total.toLocaleString()}`;
    document.getElementById('stat-high-name').innerText = maxCust.name;
    document.getElementById('stat-high-product').innerText = `Purchased:
${maxOrder.productName}`;

    // Most Active Customer (Counting grouped orders)
    const counts = {};
    orders.forEach(o => counts[o.cust_id] = (counts[o.cust_id] || 0) + 1);

    let topCustId = null;
    let maxCount = 0;
    for (const id in counts) {
      if (counts[id] > maxCount) {
```

```
            maxCount = counts[id];
            topCustId = parseInt(id);
        }
    }

    const activeCust = customers.find(c => c.id === topCustId);
    document.getElementById('stat-active-cust').innerText = activeCust.name;
    document.getElementById('stat-active-count').innerText = `${maxCount} Completed Orders`;
  }

  window.onload = init;
</script>
</body>
</html>
```

Output:





Result :

The system successfully combines data from multiple tables (Customers, Orders, and Products) to present a complete order history. It also calculates and displays the total number of order.